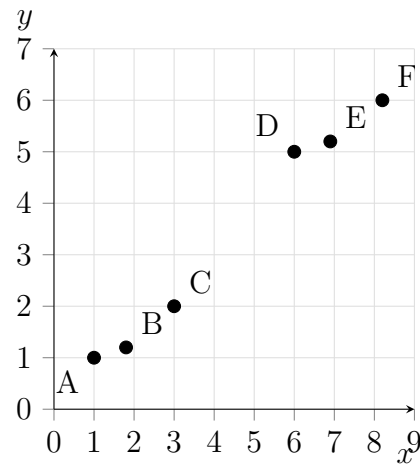


Exercise 8 Additional Practice

- Consider the dataset of 6 points given below. Perform an agglomerative clustering using Single/MIN linkage with Euclidean distance and create the dendrogram.

	x	y
A	1.0	1.0
B	1.8	1.2
C	3.0	2.0
D	6.0	5.0
E	6.9	5.2
F	8.2	6.0



The following Euclidean distance matrix is intentionally incomplete. Distances are rounded to two decimals. The upper triangle is left blank because the matrix is symmetric; entries marked “?” must be computed from the coordinates.

	A	B	C	D	E	F
A	0					
B	?	0				
C	2.24	?	0			
D	6.40	?	?	0		
E	?	6.48	5.04	?	0	
F	8.77	8.00	?	2.42	?	0

- For $C_1 = \{A, B, C\}$ and $C_2 = \{D, E, F\}$, compute the distance between C_1 and C_2 using:
 - Complete/MAX linkage
 - Group Average linkage
 - Centroid linkage
 - Ward's method, interpreted as increase in total SSE

Solutions

Solution 1: Single/MIN linkage

The missing distances needed for the clustering are:

$$\begin{aligned} d(A, B) &= 0.82, & d(B, C) &= 1.44, & d(D, E) &= 0.92, \\ d(E, F) &= 1.53, & d(C, D) &= 4.24. \end{aligned}$$

Using Single/MIN linkage, the distance between two clusters is the smallest cross-cluster point distance.

Step	Merge	Height
1	A and B	0.82
2	D and E	0.92
3	$\{A, B\}$ and C	1.44
4	$\{D, E\}$ and F	1.53
5	$\{A, B, C\}$ and $\{D, E, F\}$	4.24

Therefore the single-linkage hierarchy is

$$((A, B), C) \quad \text{and} \quad ((D, E), F),$$

followed by the final merge of the two large clusters at height 4.24.

Solution 2: Distances between $C_1 = \{A, B, C\}$ and $C_2 = \{D, E, F\}$

The cross-cluster distances are:

A, D	A, E	A, F	B, D	B, E	B, F	C, D	C, E	C, F
6.40	7.24	8.77	5.66	6.48	8.00	4.24	5.04	6.56

Complete/MAX linkage:

$$d_{\max}(C_1, C_2) = \max(6.40, 7.24, 8.77, 5.66, 6.48, 8.00, 4.24, 5.04, 6.56) = 8.77.$$

Group Average linkage:

$$d_{\text{avg}}(C_1, C_2) = \frac{6.40 + 7.24 + 8.77 + 5.66 + 6.48 + 8.00 + 4.24 + 5.04 + 6.56}{9} = 6.49.$$

Centroid linkage:

$$\mu_1 = \left(\frac{1 + 1.8 + 3}{3}, \frac{1 + 1.2 + 2}{3} \right) = (1.93, 1.40),$$

$$\mu_2 = \left(\frac{6 + 6.9 + 8.2}{3}, \frac{5 + 5.2 + 6}{3} \right) = (7.03, 5.40).$$

Thus

$$d_{\text{centroid}}(C_1, C_2) = \sqrt{(7.03 - 1.93)^2 + (5.40 - 1.40)^2} = 6.48.$$

Ward's method through SSE:

For Ward's method, compute the increase in total SSE after merging the two clusters.

$$SSE(C_1) = 2.59, \quad SSE(C_2) = 3.01, \quad SSE(C_1) + SSE(C_2) = 5.59.$$

After merging all six points into $C_1 \cup C_2$, the centroid is

$$\mu_{12} = \left(\frac{26.9}{6}, \frac{20.4}{6} \right) = (4.48, 3.40),$$

and

$$SSE(C_1 \cup C_2) = 68.61.$$

Therefore the Ward merge cost is

$$\Delta SSE = 68.61 - 5.59 = 63.02.$$

Method	Distance or merge cost
Complete/MAX linkage	8.77
Group Average linkage	6.49
Centroid linkage	6.48
Ward's method, increase in SSE	63.02